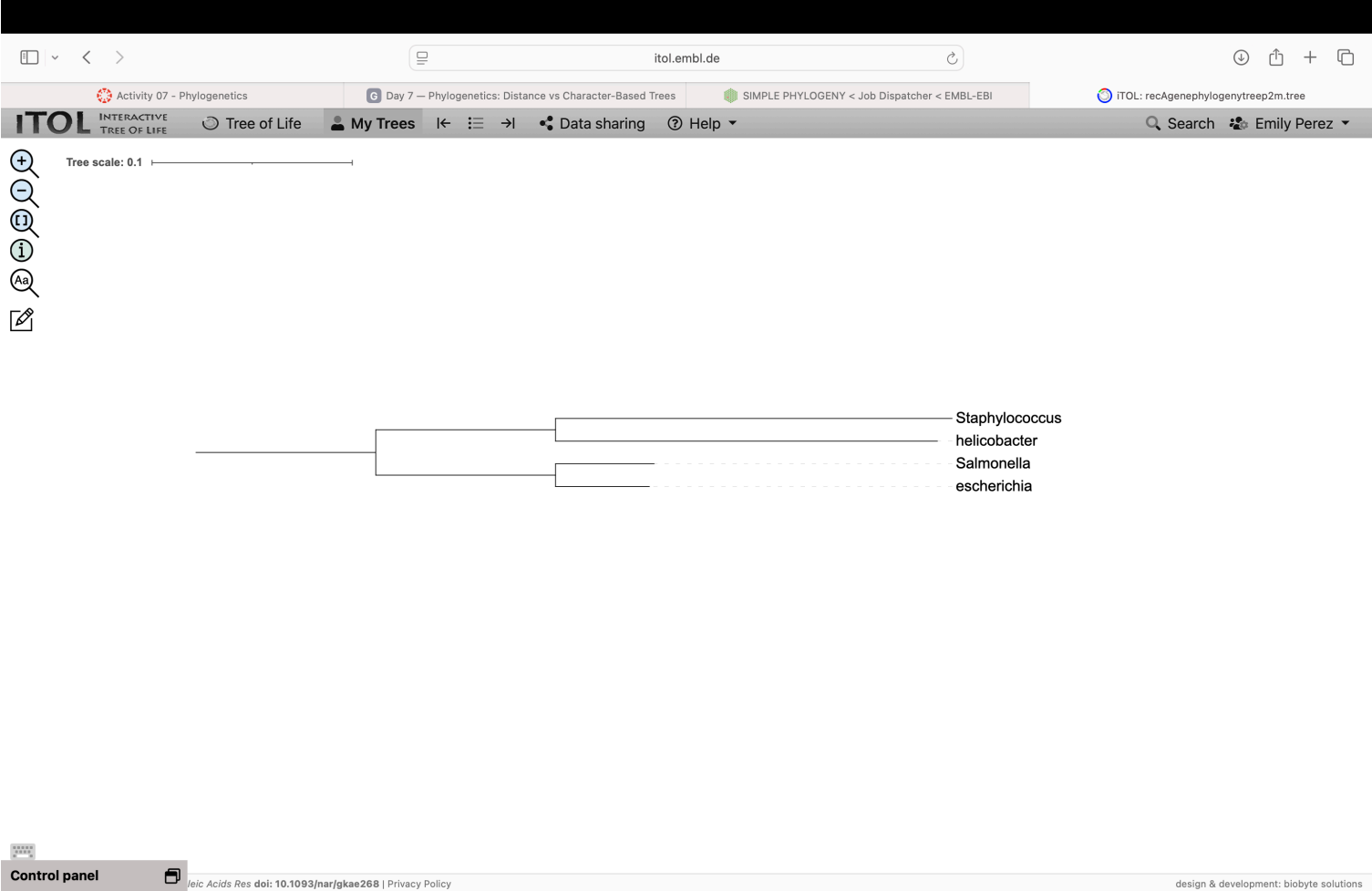


Part 1 — FastME (distance-based)

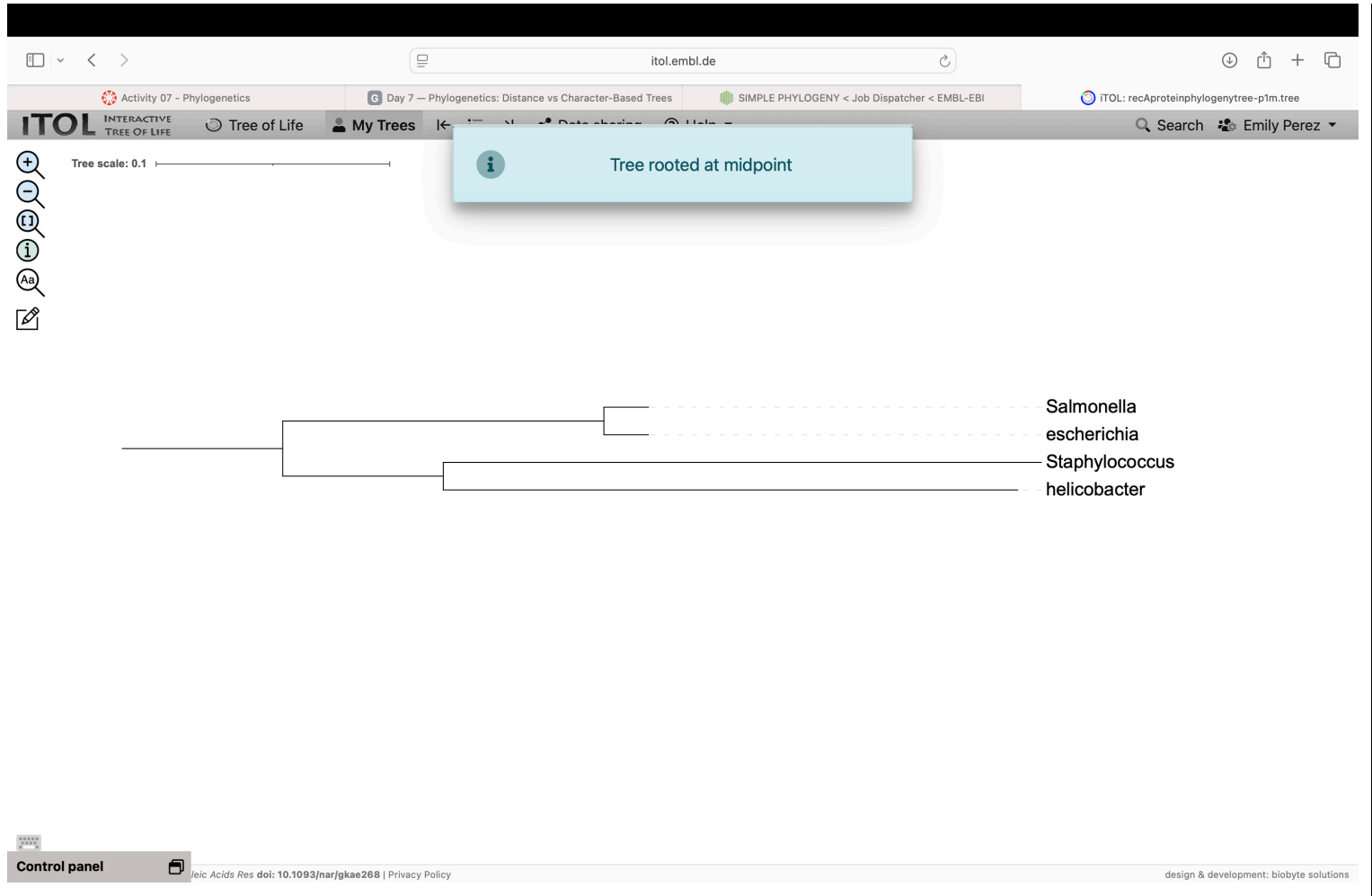
A1: The DNA and protein trees do have the same topology. They are basically the same except that the protein put Salmonella and Escherichia on top and the DNA but them at the bottom. The DNA tree has a longer line for Salmonella and Escherichia than the protein tree

A2: This website didn't do the bootstrap values.

FastME DNA tree:



FastME Protein tree:



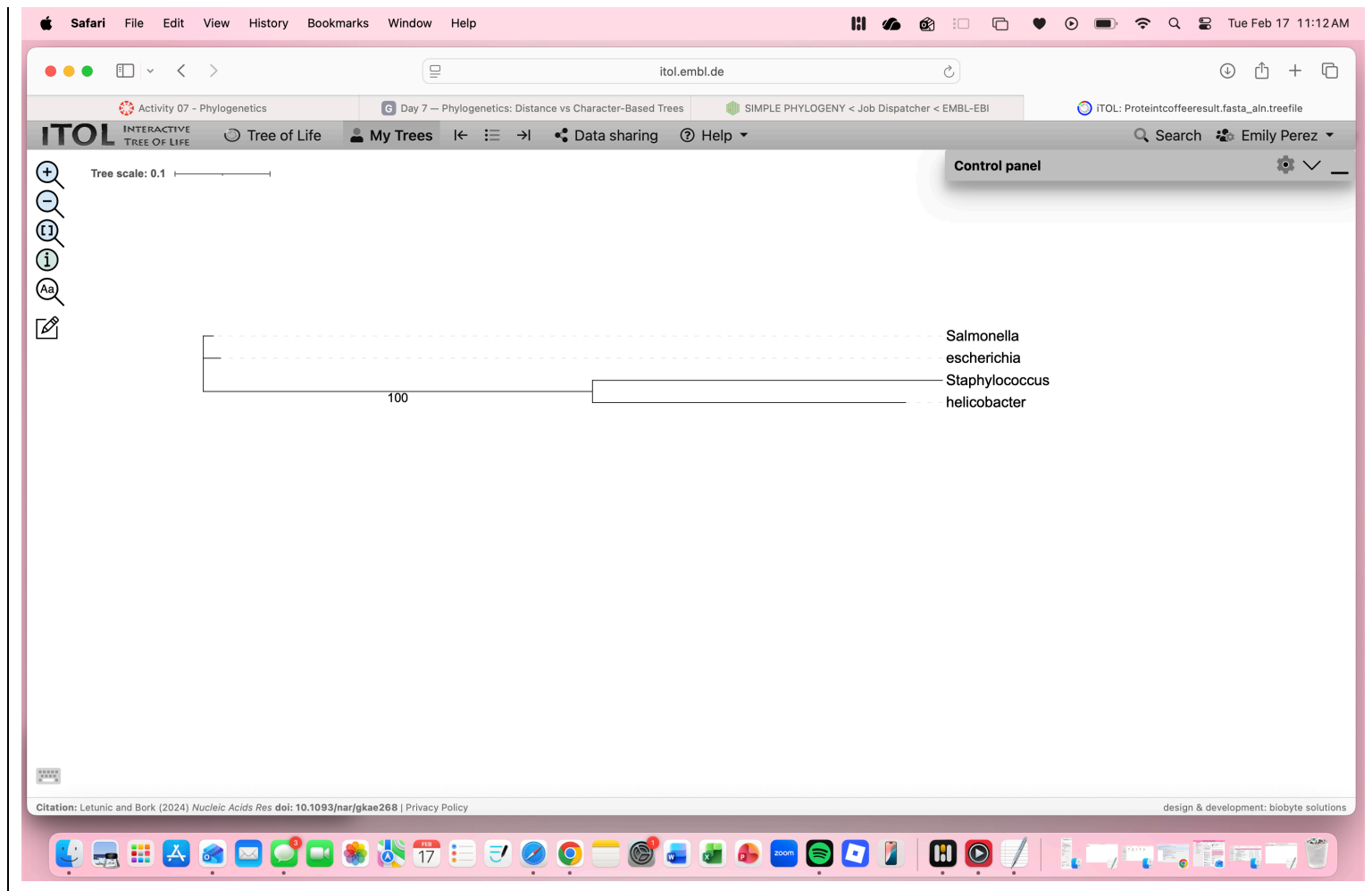
Part 2 – PhyML-SMS (maximum likelihood)

B1: They do have the same topology. Where staphylococcus and helicobacter are connected and salmonella and escherichia are connected from a different root.

B2: The branches between salmonella and escherichia are smaller and same with the branch with helicobacter. Assumptions might explain this difference because some programs might read the data differently than others.

B3: I cannot compare values.

ML Protein tree (PhyML-SMS):



Part 3 – Rooting (concepts)

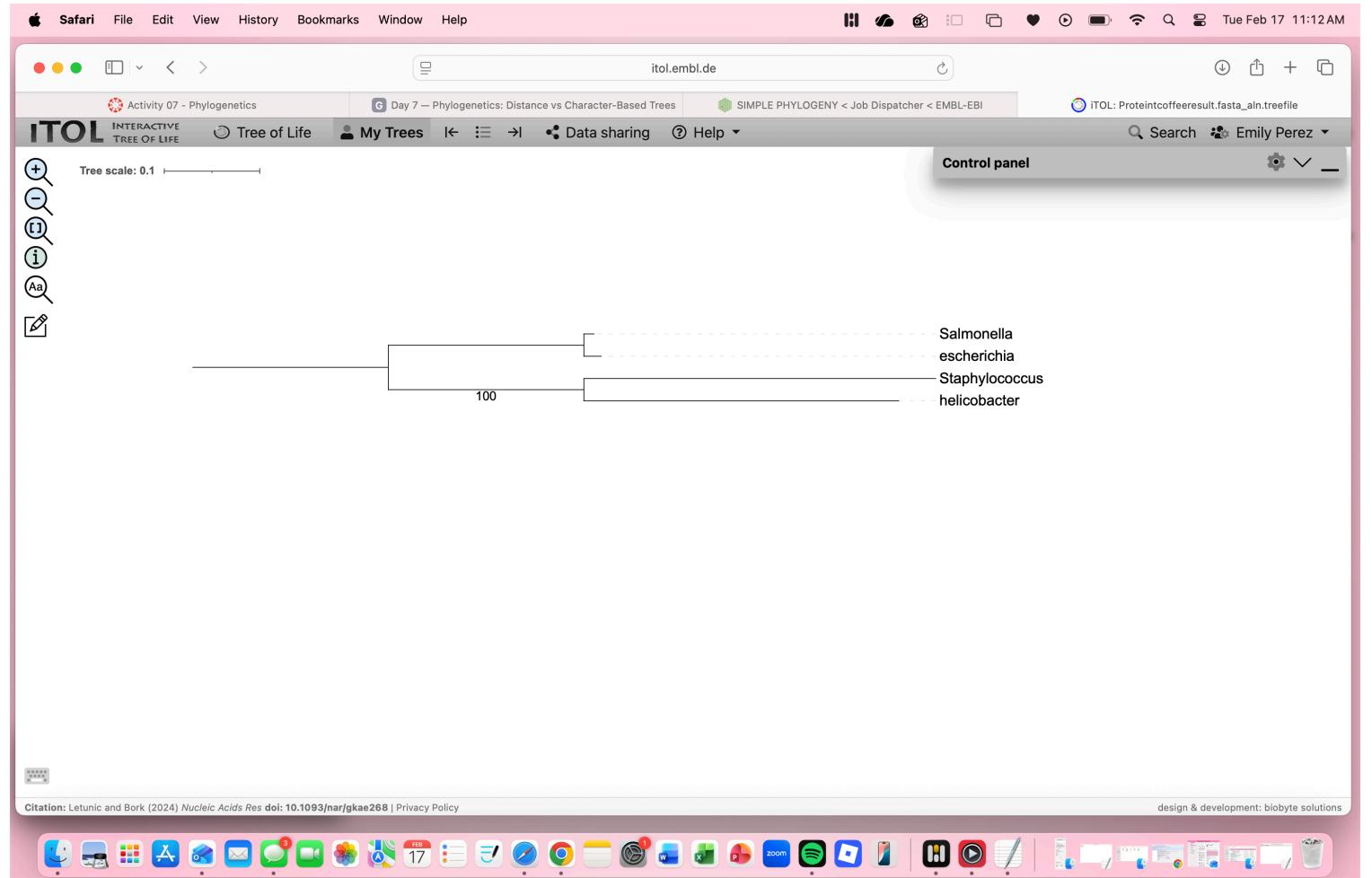
C1: Midpoint rooting changes it so that there is a common ancestor. It all branches from one root to the 4.

C2: Midpoint rooting changes it the same as the FastME tree. There are just smaller branches.

C3: An assumption that midpoint rooting might make is that it assumes that the two most divergent branches have evolved at roughly the same speed and are from a true ancestor. This can be misleading because it is simply not true.

Part 4 – Synthesis + rooted ML

Midpoint-rooted ML Protein tree:



D1: I believe that distance vs ML changed my interpretation the most. ML is more precise in my opinion. It showed a greater difference between the species.

D2: This activity reinforces that phylogenetic trees are hypotheses because as we saw, the same alignment can lead to different trees depending on the website.

